

An ad for ADT Security Systems says,

“When you go on vacation, burglars go to work [...] According to FBI statistics, over 25% of home burglaries occur between Memorial Day to Labor Day.”

Do the data in the ad support the claim that burglars are more likely to go to work during the time between Memorial Day to Labor Day? Please explain your answer. **(6 Points)**

Note: You can assume that “over 25%” means only slightly over. Had it been much over, say closer to 30%, then the marketers would have said so.

Note: Memorial Day is observed on the last Monday of May and Labor Day is observed on the first Monday of September.

No, because the time between Memorial Day to Labor Day is around 100 days, and that is roughly 27% of a year, and if burglars were spread uniformly across a year it'd be expected that 27% of burglaries would happen from Memorial Day to Labor Day. $25\% < 27\%$, so the claim that burglars are more likely to happen between Memorial Day to Labor Day is false.

Question 1. The data were gathered by the following procedure, reported in the study. “Between January and June 1998, parents of children aged 2-16 years [...] that were seen as outpatients in a university pediatric ophthalmology clinic completed a questionnaire on the child’s light exposure both at present and before the age of 2 years.” Was this study observational, or was it a controlled experiment? Explain. **(5 Points)**

observational, there was no treatment provided, and people weren’t selected at random.

Question 2. The study found that of the children who slept with a room light on before the age of 2, 55% were myopic. Of the children who slept with a night light on before the age of 2, 34% were myopic. Of the children who slept in the dark before the age of 2, 10% were myopic. The study concluded the following: “The prevalence of myopia [...] during childhood was strongly associated with ambient light exposure during sleep at night in the first two years after birth.”

Do the data support this statement? Why or why not? You may interpret “strongly” in any reasonable qualitative way. **(5 Points)**

Yes, there is a strong association. There is an increase of myopic children as light exposure increases. The number of children who were myopic jumped from 10% where no light was present to 34% with a night light, to 55% where the highest amount of room light was present.

Question 3. On May 13, 1999, CNN reported the results of this study under the headline, “Night light may lead to nearsightedness.” Does the conclusion of the study claim that night light causes nearsightedness? (5 Points)

No, it does not directly claim that night light is a causation for nearsightedness.

Question 4. The final paragraph of the CNN report said that “several eye specialists” had pointed out that the study should have accounted for heredity.

Myopia is passed down from parents to children. Myopic parents are more likely to have myopic children, and may also be more likely to leave lights on habitually (since the parents have poor vision). In what way does the knowledge of this possible genetic link affect how we interpret the data from the study? Explain. **(5 Points)**

Myopic parents are genetically able to have myopic kids and are likely to leave lights on, so this could explain the association between myopic children and light exposure.

